

The Standing Fixity of Physical Constants

Admissibility Constraints on Fine-Tuning, Anthropic Selection, and Multiverse Variation

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Abstract

Explanations of physical constants frequently appeal to fine-tuning, anthropic selection, landscape variation, or multiverse sampling. These strategies usually presuppose that constants range over a meaningful parameter space of alternative values and that this space already supports comparison, probability, typicality, and cross-realization reference. This paper challenges that presupposition. A physical constant, when it is treated as standing-bearing inside a fixed physical domain, is not a freely variable same-object parameter. It is an invariant role in the standing regime that permits physical quantities to be stably referred to, transported, compared, measured, and used across admissible redescriptions.

The claim is deliberately narrow. The paper neither derives the numerical values of constants nor asserts metaphysical necessity across all possible worlds. It is also compatible with renormalization-group running, empirical calibration, and effective-field-theory parameters. It establishes a fixed-domain result: within an admissibility-bearing physical domain, any attempted variation of a standing-bearing constant is either skin-level reparameterization, envelope or boundary input, RG transport under licensed continuation, theory/domain change, or a same-domain value sweep that fails to preserve the standing conditions of the target. The reason is not stipulative: a parameter that lacks standing-preserving empirical transport does not yet function as a physical constant for that domain. Fine-tuning and anthropic arguments remain possible only after an admissible ensemble, an explicitly specified measure with declared authority, and a cross-domain witness ledger have been supplied; they do not themselves supply the standing structure whose absence is at issue. The result reclassifies the explanatory role of constants and identifies the structure required before a parameter space can carry canonical explanatory authority: a standing-preserving metric, measure, selector, and identity-transport rule must be supplied rather than inferred from notation alone.

Orientation. The argument has two complementary parts. First, a raw space of alternative constant-values is not automatically a physical object. Second, constant-fixity reappears at the correct level: standing-bearing constants are fixed relative to the admissible domain they help constitute. Apparent counterexamples are sorted by role: unit and chart changes are skin, running couplings are continuation data, empirical values may be envelope inputs, and genuine changes of constant-role are changes of domain or theory rather than free variations of the same standing object.

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1 Introduction

Physical constants occupy a peculiar position in modern theoretical physics. They appear in equations as fixed numerical inputs, but the equations themselves often do not explain why those inputs have the values they do. This has encouraged three familiar explanatory strategies. First, one may treat the values as fine-tuned and ask for the probability that a randomly sampled universe would land in the life-permitting range. Second, one may appeal to anthropic selection: observers find themselves only in domains compatible with observers. Third, one may posit a landscape or multiverse in which different realizations instantiate different low-energy constants, so that our constants are selected from a broader ensemble.

Despite their differences, these strategies share a structural presupposition. They treat the relevant constants as if they ranged over a parameter space of alternatives that is already meaningful enough to support comparison, counting, weighting, typicality, and reference. The parameter space is often drawn before the physical admissibility of the comparison has been secured. A coordinate interval may be mistaken for a physical ensemble; a numerical knob for a standing-preserving degree of freedom; a possible equation with altered symbols for a same-domain physical alternative.

This paper argues that this presupposition requires justification. The central distinction is between *admissible redescription* and *non-standing-preserving variation*. An admissible redescription changes notation, coordinates, units, gauge, scheme, or other presentational features while preserving the standing of the physical content. Non-standing-preserving variation changes the role by which the theory fixes physical reference while still pretending to compare the same object. The explanatory puzzle changes once these categories are separated.

The thesis is:

Within a fixed admissibility-bearing physical domain, a standing-bearing physical constant cannot be treated as a freely variable same-object parameter. Its apparent alternative-values space is admissible only after role preservation, witness transport, boundary status, and comparison measure have been independently supplied.

Although the paper uses the language of admissibility and standing, the constraint is not proprietary to AASC. Any physical framework that permits stable cross-context empirical transport, parameter identification, calibration, comparison, and theory application must enforce structurally analogous invariance conditions. AASC supplies a compact vocabulary for naming the constraint; the physical requirement is stable parameter identity under licensed use.

The thesis is not a numerical derivation of constants. It is a constraint on explanatory form. The paper does not claim that no different theory could have different constants, that no effective parameter may run, that no multiverse model may define an ensemble, or that constants are logically necessary in all possible worlds. It claims that a same-domain explanatory sweep over constant-values is not yet licensed unless the sweep preserves the standing regime required for physical reference.

The argument uses four pieces of role discipline. First, standing-bearing physical reference requires stable identity across admissible redescriptions. Second, the anchor/tensor/skin decomposition separates admissibility preconditions, invariant physical load, and presentational skin. Third, the AMetric boundary withholds canonically privileged metric, measure, selector, typicality, and representative authority until admissible structure is fixed. Fourth, the null-regime and constraint-bundle material explains why a vacuous, arbitrary, or lower selector cannot govern constants from outside the admissible domain.

The result is a formalized version of the Standing Fixity claim. Constants are not explained by sampling from a pre-given space of alternative values. The proposed space must first be shown to

be a standing-preserving object. If it is only a chart, the variation is skin. If it changes observable compatibility or domain governance, it is theory/domain change. If it demands a probability distribution before admissible measure has been fixed, it introduces selector structure not supplied by the fixed domain. If it appeals to anthropic selection before a cross-domain ensemble has standing, the selection step depends on comparison structure that has not yet been supplied.

1.1 Contributions

The paper makes twelve contributions.

- (1) It replaces the informal statement “constants are fixed” with a fixed-domain comparison class for constant-variation claims.
- (2) It distinguishes dimensionful unit-sensitive quantities, raw chart-numeric parameters, effective/running parameters, envelope inputs, and standing-bearing invariant load.
- (3) It establishes a theorem ladder: empirical transport, skin-variation, tensor-load, no same-domain free constant, and standing fixity.
- (4) It classifies apparent constant-variation routes and identifies the same-domain value sweep as the route excluded for standing-bearing constants.
- (5) It establishes a no-canonical-measure result for fine-tuning: before standing is fixed, the putative constant-space has no canonically privileged standing-preserving metric, probability distribution, typicality order, or representative unless extra ensemble structure is declared.
- (6) It separates same-standing ensembles from cross-standing ensembles, so that multiverse claims can be evaluated without either dismissing them wholesale or granting them comparison structure not yet supplied by the model.
- (7) It gives an explicit RG-transport proposition separating lawful scale continuation from arbitrary same-domain reassignment.
- (8) It adds a constant-continuation locus criterion explaining what persists through lawful re-description, RG transport, matching, and family-level comparison.
- (9) It distinguishes formal constant-family notation, inherited standing-bearing constant multiplicity, and explicitly gated target-domain families.
- (10) It preserves controlled parameter-family scans as admissible modeling tools when they are performed inside explicitly governed target domains.
- (11) It incorporates transport-closure and finite-multiplicity discipline for continuous parameter families and replicated sectors without turning the Fixity theorem into a general uniqueness thesis.
- (12) It supplies a compact fine-structure-constant illustration showing how unit change, scheme change, RG continuation, target-domain comparison, and value-sweeps without preservation structure are classified differently.

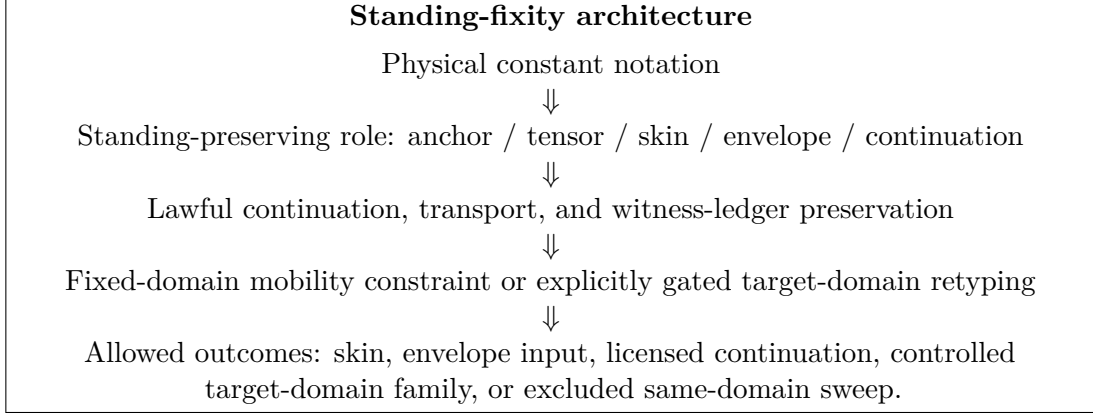


Figure 1: The Fixity theorem operates by role typing before value-space explanation. It does not deny controlled scans; it denies inherited standing-bearing authority to a value space whose comparison structure has not been supplied.

2 Scope, Non-Claims, and the Fixed-Domain Target

Because claims about constants are easily overstated, the scope of the paper is fixed at the outset.

2.1 What is not being claimed

The paper does not claim any of the following:

- (N1) **No numerical derivation.** The paper does not derive the measured values of c , $\hbar$, G , α , Yukawa couplings, mixing angles, mass ratios, or cosmological parameters.
- (N2) **No metaphysical necessity.** The result is not a claim of logical or metaphysical necessity across all possible worlds. It is a fixed-domain admissibility result.
- (N3) **Compatibility with RG running.** Running couplings and scale-dependent effective parameters are not denied. They are classified as licensed continuation or scheme-dependent chart data unless invariant trajectory content is explicitly fixed.
- (N4) **Compatibility with empirical calibration.** Measured constants may be empirical boundary or envelope inputs. The paper does not make observation unnecessary.
- (N5) **Compatibility with effective theories.** Effective-field-theory coefficients can be valid within a bounded envelope. The paper constrains only the interpretation of such coefficients when they are transported beyond the envelope that gives them standing.
- (N6) **Compatibility with alternative theories.** Different theories may have different parameter lists or constants. Such cases are theory/domain changes, not free same-domain variations of an already fixed constant.
- (N7) **Compatibility with ensemble models.** A multiverse, landscape, or statistical ensemble may be posited. But the ensemble must supply its own admissible comparison structure, measure, and witness ledger; these are not supplied merely by writing a parameter interval.

The analysis does not prohibit exploratory parameter studies, phenomenological scans, Bayesian sensitivity analysis, effective-field-theory modeling, beyond-Standard-Model construction, UV-completion searches, or model-family construction. Such studies may be useful inside declared model families. The constraint is interpretive: a scan becomes an explanation of standing-bearing constants only when the relevant admissibility envelope, witness ledger, transport rules, and measure authority are explicitly supplied. Effective field theories remain fully admissible within their operational domains; the present argument concerns only the interpretive status of parameter variation across fixed standing-bearing domains.

Remark 2.1 (Controlled parameter-family modeling). *Effective parameter families, sensitivity analyses, Wilson-coefficient scans, phenomenological fits, and explicit model-family searches remain admissible modeling tools when they are conducted inside a governed target domain with declared object-class, envelope, witness ledger, transport rules, and measure conventions. The Fixity theorem targets only the promotion of an inherited value-family into an explanation of standing-bearing constants before that governing structure has been supplied.*

Definition 2.2 (Formal, inherited, and explicitly gated constant-families). *A formal constant-family is a written family of symbols, equations, or assignments, such as $C \in I$ or $\{C_i\}_{i \in I}$, without by itself carrying standing-bearing authority. An inherited constant-family is a formal family presented as if its members automatically inherit the standing, witness ledger, transport, and comparison authority of an original domain. An explicitly gated target-domain constant-family is a newly typed domain in which membership, object-class, witness structure, transport, quotient identity, boundary status, and any measure or selector have been declared and subjected to admissibility constraints.*

Proposition 2.3 (Target-domain retyping rule). *A constant-family that supplies its own admissibility gate, transport structure, witness ledger, quotient identity, and envelope is not an inherited same-domain value sweep. It is a newly typed target-domain construction. Its use may be legitimate, but any explanatory force belongs to that new governed construction rather than to multiplicity inherited from the original constant notation.*

Proof. The added gate, transport, witness ledger, quotient, and envelope determine a new comparison target. If the new structure is absent, the family does not inherit standing-bearing authority from notation alone. If the structure is present, the comparison is no longer a free same-domain sweep of the original standing-bearing constant; it is a target-domain construction with its own admissibility obligations. $\square$

2.2 Fixed domains and constant comparison

The argument requires a precise target. A constant cannot be varied “in the same theory” unless the relevant sense of sameness has been fixed.

Definition 2.4 (Fixed admissibility domain). *A fixed admissibility domain is a comparison-relative structure*

$$D = (\mathcal{T}, \mathcal{A}, \mathcal{W}, \mathcal{R}, \mathcal{Q}, \mathcal{E}),$$

where $\mathcal{T}$ is the operative theory or theory-family, $\mathcal{A}$ is the admissibility envelope, $\mathcal{W}$ is the witness ledger by which quantities are empirically identified, $\mathcal{R}$ is the lawful redescription and transport structure, $\mathcal{Q}$ is the standing quotient identifying equivalent presentations, and $\mathcal{E}$ records boundary, calibration, cutoff, and envelope data. The domain is fixed for a proposed comparison only when witness compatibility, lawful transport, standing-preserving redescription, invariant object identification, and envelope continuity are preserved. A comparison that changes any of these components may

still be physically meaningful, but it is no longer a same-domain comparison unless an additional bridge supplies the missing transport.

Remark 2.5 (Operational status of the definition). *Definition 2.4 does not add a new metaphysical postulate. It collects the conditions already needed for ordinary physical use of a parameter: a measurement in one context must be identifiable with an equation-entry in another context, and both must remain connected by licensed transport rules.*

A constant-variation claim remains in the same target only if it stays inside a fixed admissibility domain and preserves the following components.

Definition 2.6 (Fixed-domain constant comparison class). *A proposed variation, redescription, or rival construction remains in the same constant-comparison class for (D, C) , where D is fixed in the sense of Definition 2.4, only if it preserves all of the following simultaneously:*

- (C1) **Admissibility anchors:** bivalence/fail-closure, standing conservation, no unsupported scope transport, and no new primitives.
- (C2) **Object-class:** the target is not equivocally shifted between a raw numerical label, a unit convention, a chart coordinate, a running effective parameter, an observable relation, and a standing-bearing invariant.
- (C3) **Standing quotient:** admissible redescriptions identify the same physical content by the same quotient relation.
- (C4) **Witness ledger:** empirical or observational records used to individuate the quantity remain comparable under the same evidence-role.
- (C5) **Tensor package:** invariant compatibility data, symmetry constraints, representation constraints, locality/causality constraints, and explicitly fixed normalization conventions remain fixed.
- (C6) **Continuation and transport:** lawful time evolution, scale evolution, RG flow, gauge/scheme transformations, and cross-context translation rules remain licensed.
- (C7) **Boundary/envelope status:** boundary conditions, effective cutoffs, calibration inputs, and domain envelopes are not silently changed.
- (C8) **Role carrier:** the anchor/tensor/skin role assignment is evaluated on the same carrier.

Changing any one of these components is a scope change. It may be a legitimate new theory, new model, new envelope, or new empirical problem, but it is not a same-domain variation of the original standing-bearing constant.

Lemma 2.7 (Scope-shift isolation). *Any purported counterexample to standing fixity that changes one of the components in Definition 2.6 either exits the target comparison class or performs unsupported scope transport if it continues to present itself as same-domain.*

Proof. The target is individuated by the preserved components. If a maneuver changes the object-class, quotient, witness ledger, tensor package, continuation rules, boundary status, or role carrier, then the object under audit has changed. To continue treating the changed object as the same standing object requires a transport rule between domains. If such a rule is supplied and licensed, the maneuver is no longer a free same-domain value variation; it is a cross-domain construction with additional structure. If no such rule is supplied, the maneuver transports standing without a witness, violating standing conservation and no-new-primitives. $\square$

Proposition 2.8 (Inherited constant-multiplicity non-inheritance). *Formal multiplicity of constant-values does not by itself inherit standing-bearing authority. A family of alternatives for a standing-bearing constant either (i) remains formal or bookkeeping, (ii) is retyped as an explicitly gated target-domain construction, (iii) changes the original domain or envelope, or (iv) lacks the transport and witness structure required for admissible same-domain comparison.*

Proof. A standing-bearing comparison requires the components in Definition 2.6. A written multiplicity supplies none of them merely by being indexed. If the components are added, Proposition 2.3 applies. If they are not added, the family has no inherited standing-bearing authority. If the components are altered rather than preserved, the comparison exits the original fixed domain. These alternatives exhaust the ways a constant-family can be used without silently importing a selector or standing-transfer rule. $\square$

2.3 Why the fixed-domain restriction matters

The fixed-domain restriction is not a protective clause. It is the condition under which the question has a determinate target. Without it, a question such as “Why does this constant have this value?” equivocates between at least five different objects:

- (i) a unit convention;
- (ii) a coordinate value in a chosen parameterization;
- (iii) a measured boundary input;
- (iv) an effective value at a scale and scheme;
- (v) an invariant physical relation whose alteration changes standing-bearing compatibility.

Only the last object is a candidate for standing-fixity. The others may be important, but they do not support a same-domain fine-tuning sweep unless additional admissible structure has been fixed.

3 Standing, Admissibility, and Physical Reference

This section records the minimal formal vocabulary used later.

Definition 3.1 (Admissible redescription). *An admissible redescription of a physical domain is a representational transformation that preserves:*

- (R1) *the identity of the standing object under discussion;*
- (R2) *all standing-bearing invariant compatibility relations;*
- (R3) *the licensed witness ledger and continuation rules;*
- (R4) *the anchor/tensor/skin role assignment.*

Differences confined to notation, coordinate choice, units, chart convention, gauge presentation, scheme presentation, or other non-load-bearing expression are skin-level differences.

Definition 3.2 (Standing identity). *A standing identity is an equivalence class of presentations under admissible redescription. Two presentations stand for the same physical object only when they lie in the same admissibility-preserving equivalence class.*

Definition 3.3 (Standing-bearing quantity). *A quantity q is standing-bearing in $(\mathcal{T}, \mathcal{A})$ iff:*

- (S1) q is invariant under all admissible redescription that preserve the target object-class; and*
- (S2) changing the role or value assigned to q changes tensor compatibility, witness compatibility, or standing identity for the fixed domain.*

If q varies under admissible redescription while all standing-bearing compatibility is preserved, q is not standing-bearing as a raw value. It is a chart, unit, or skin-level coordinate of some other invariant content.

Standing-bearing invariants, in this usage, are not merely formal quantities left unchanged by a transformation. They are invariants required for stable empirical identification and lawful transport across admissible observational contexts.

Definition 3.4 (Standing-bearing constant). *A standing-bearing physical constant is a quantity C used by $(\mathcal{T}, \mathcal{A})$ as fixed load in the standing structure of the domain. It may appear numerically in equations, but its standing role is not exhausted by a printed number: it is the role by which admissible reference, comparison, continuation, and witness compatibility remain stable.*

Remark 3.5 (Printed parameters versus physical constants). *A parameter printed in a Lagrangian, Hamiltonian, equation of state, effective action, or phenomenological model is not automatically a standing-bearing constant. It may be a coordinate, normalization, fit coefficient, running value, boundary input, or envelope-local parameter. Role analysis is required before the Fixity theorem applies.*

Definition 3.6 (Constant-continuation locus). *A constant-continuation locus for a standing-bearing constant C is the admissibility-bearing identity by which C is retained as the same structural role across lawful redescription, calibration, scheme change, matching, and transport. Operationally, it is what permits successive presentations of C to count as the same standing-bearing constant rather than as unrelated coordinates, fresh boundary inputs, or a newly defined parameter in another target domain. It is not the isolated printed number. It is the preserved role linking object-class, witness ledger, standing quotient, boundary status, and transport behavior.*

Proposition 3.7 (Constant-continuation criterion). *A proposed variation of C remains in the same standing-fixity target only if it preserves the constant-continuation locus of C . If the locus is preserved, the variation is skin, licensed continuation, or envelope-local modeling. If the locus is not preserved, the proposal is a target-domain construction, a domain change, or an unsupported same-domain sweep.*

Proof. The continuation locus records what must remain fixed for C to be treated as the same physical constant through use. Unit, chart, gauge, scheme, or RG presentations may alter numbers while preserving that locus. A change that alters the locus changes the identity of the standing-bearing role and therefore cannot be a free same-domain variation of the same constant unless an independently gated target domain supplies the missing comparison structure. $\square$

Proposition 3.8 (Empirical transport condition). *A parameter that lacks standing-preserving empirical transport does not function as a physical constant for a fixed domain. More precisely, if no invariant identification relation carries the parameter across admissible observers, times, calibrations, experiments, and lawful redescription, then the parameter is at most a local coordinate, fitted label, boundary tag, or chart-specific entry. It does not support same-domain comparison, probability, or fine-tuning claims as a physical constant.*

Proof. A physical constant is used across contexts: it is calibrated in one setting, inserted into equations in another, compared across observers, and retained through lawful redescription. This use requires an invariant identification relation. Without such a relation, there is no stable fact of the matter that the same parameter is being tracked rather than a different chart value, convention, fit coefficient, or domain-local fitted coefficient. The missing transport is not supplied by a probability distribution over printed numbers, because that distribution already presupposes that the numbers name comparable objects. Hence standing-preserving empirical transport is a minimal condition for physical-constant status rather than an optional philosophical add-on. $\square$

Corollary 3.9 (Standing is not stipulative). *The standing condition is not an added metaphysical preference. It is the operational condition under which empirical records, calibrations, and redescription-equivalent formulations continue to identify one physical parameter rather than unrelated local symbols.*

Proof. Remove standing-preserving transport and a value measured in one context no longer has a licensed identity relation to a value written in another. Calibration, cross-context comparison, probability over alternatives, and reuse of a parameter in equations all presuppose this relation. The condition is therefore required by the practice of physical parameter use before any further metaphysical interpretation is added. $\square$

Lemma 3.10 (Standing-admissibility linkage). *On a fixed domain, standing and admissibility cannot be separated by a second same-domain classifier. If a transformation preserves admissibility for the same object-class, it preserves standing; if it changes standing-bearing identity or compatibility, it is not an admissible redescription of that object.*

Proof. A same-domain admissibility verdict must be issued about a stable target. If the target changes while the verdict is treated as same-object, the verdict equivocates. Conversely, if the target is preserved and all load-bearing compatibility data are preserved, the transformation has done only representational work. A second classifier that tries to decide, on the same scope, which of two standing-incompatible presentations is admissible would need authority not supplied by the fixed domain. It either imports a new primitive selector or changes the domain. $\square$

Corollary 3.11 (No repair by redescription). *A standing failure caused by changing a standing-bearing constant cannot be repaired by later relabeling the changed theory as a redescription of the original one.*

Proof. By Lemma 3.10, admissible redescription preserves standing. A later relabeling that restores sameness after standing-bearing compatibility has changed would create standing rather than preserve it. That is not redescription; it is unsupported repair or theory change. $\square$

4 Anchor/Tensor/Skin Role Discipline for Constants

The same symbol may play different roles in different contexts. The role distinction used here is anchor/tensor/skin.

Definition 4.1 (Anchor, tensor, and skin). *Within a fixed physical domain:*

(A1) *An anchor is a precondition for admissible reference or evaluation. Examples include standing conservation, no unsupported scope transport, fail-closure, and fixed witness-role.*

- (A2) *A tensor is load-bearing invariant content: compatibility relations, standing-preserving constraints, observable relations, fixed quotient data, or other physical content not removable by admissible redescription.*
- (A3) *Skin is presentation: notation, coordinate choice, unit normalization, chart selection, gauge display, scheme convention, or other expression that carries no independent standing authority.*

The distinction is not cosmetic. Most confusion about constants comes from promoting skin to tensor or treating tensor change as if it were skin.

Anchor	admissibility conditions, law structure, witness authority, identity persistence, and domain continuity
Tensor	invariant physical load: conserved relations, observable compatibility, coupling or mixing structure when role-bearing, fixed quotient data
Skin	coordinates, units, gauge display, renormalization scheme, notation, normalization, and other chart-level representations

The diagram is not a hierarchy of metaphysical reality; it is a hierarchy of explanatory authority. Skin can display tensor load, but it does not select it. Tensor load can constrain admissible presentation, but it is evaluated only against the anchors that permit stable reference. Anchors supply the conditions under which tensor and skin roles are even evaluable within a fixed domain.

Lemma 4.2 (Role exhaustion). *Within a fixed constant-comparison class, every purported constant-variation is exactly one of the following: anchor change, tensor change, skin change, envelope/boundary change, licensed continuation, or same-domain variation without preservation structure.*

Proof. By Definitions 2.4 and 2.6, the fixed target includes anchors, object-class, quotient, witness ledger, tensor package, continuation, boundary status, and role carrier. A proposed variation either preserves these data, changes one of them under a licensed rule, or changes one of them without a licensed rule. If all standing-bearing data are preserved, the variation is skin. If invariant compatibility changes, it is tensor change. If admissibility preconditions change, it is anchor change. If calibration, cutoff, boundary, or domain envelope changes, it is envelope/boundary change. If the change is produced by a licensed evolution such as RG flow, it is continuation. If none of these licenses apply while same-domain sameness is still asserted, the variation is outside the licensed comparison class. $\square$

Lemma 4.3 (Skin non-authority). *Skin does not select, justify, or generate tensor content. A coordinate, unit convention, chart, or scheme can express standing-bearing content, but it cannot make one value physically privileged unless the privilege is already fixed by tensor or anchor data.*

Proof. A skin-level difference is defined by preservation of standing-bearing content. If a skin element selects one contentful value over another, it changes admissibility-bearing outcome and is no longer skin. If the selection is not already fixed by tensor or anchor data, the skin element has been promoted into an unsupported selector. Hence skin can display content but cannot generate it. $\square$

A constant cannot be skin and standing-bearing tensor load in the same respect. It may have a skin expression, such as a number in chosen units or a coupling written in a chosen scheme, while

the invariant relation expressed through that skin carries tensor load. The distinction is therefore not between “real constants” and “unreal constants.” It is between the role-bearing content that survives admissible reparameterization and the presentation through which that content is displayed.

Lemma 4.4 (Skin-tensor exclusion). *Within a fixed constant-comparison class, a raw value cannot be both mere skin and standing-bearing tensor load in the same respect.*

Proof. If the raw value is mere skin, it can change under admissible redescription while the standing quotient, witness ledger, and tensor package are preserved. If the same raw value is standing-bearing tensor load in the same respect, changing it changes invariant compatibility. The same change therefore both preserves and fails to preserve standing-bearing content, which is not available on fixed scope. The consistent reading is that the raw value is skin while some invariant relation expressed by it may be tensor load. $\square$

Lemma 4.5 (Tensor preservation under admissible reparameterization). *An admissible reparameterization may change the symbol, coordinate, unit, gauge display, or scheme in which a constant-role is expressed, but it preserves the tensor load carried by that role.*

Proof. Admissible redescription is defined by preservation of the standing quotient, witness ledger, and load-bearing compatibility relations. If a reparameterization changed tensor load, it would change what must be preserved and would no longer be a redescription of the same standing object. Thus admissible reparameterization changes the carrier or display while preserving the invariant role-content. $\square$

Lemma 4.6 (Anchor alteration is domain alteration). *Changing an anchor that supplies admissible reference, witness authority, scope discipline, or standing conservation is not a reparameterization of the same domain. It is a change of the admissibility envelope unless an explicit transport rule to a larger domain is supplied.*

Proof. Anchors are the preconditions under which physical reference is available. A change in an anchor changes the conditions under which the target object is identifiable, testable, or comparable. Such a change may be legitimate as theory extension or domain change, but it cannot be treated as a same-domain parameter variation unless a higher-level admissible map preserves the relevant object-class and witness ledger. $\square$

5 The Standing Fixity Theorem Ladder

The Fixity theorem is most precise when stated as a ladder of smaller claims.

Lemma 5.1 (Standing-bearing quantity criterion). *A quantity is a standing-bearing quantity only if it is invariant under admissible redescription and its alteration changes standing-bearing compatibility for the fixed domain.*

Proof. If the quantity is not invariant under admissible redescription, then the same physical object can be represented with different values of the quantity while all standing-bearing content is preserved. The raw value is therefore skin. If alteration of the quantity changes no standing-bearing compatibility, it is again non-load-bearing. Conversely, if the quantity is redescription-invariant and alteration changes compatibility, it carries standing load. $\square$

Apparent constant object	Likely role	Fixity implication
Choice of units for c , $\hbar$, k_B	Skin/normalization	Numerical value can change by convention; invariant relations do not.
Raw value of a coupling at a chosen scheme/scale	Skin plus continuation coordinate	The number is not standing-bearing as raw number; invariant trajectory or observable relation may be.
Measured value of a parameter inside a bounded model	Boundary/envelope input	Fixed for the domain once calibrated, but not derived by the model alone.
Mass ratio, mixing angle, or dimensionless relation used as invariant input	Candidate tensor or envelope load	Requires role analysis; a prior over alternatives has physical explanatory status only after admissible ensemble structure is supplied.
Symmetry, anomaly, locality, or representation compatibility condition	Tensor/anchor load	Variation changes the domain or standing package.
Counterfactual sweep over all possible numerical values	Requires ensemble structure	A coordinate interval is not a physical parameter space by itself.

Table 1: Role discipline for apparent constants.

Lemma 5.2 (Skin-variation lemma). *If a purported change in a constant preserves the standing quotient, witness ledger, tensor compatibility, and continuation structure of the fixed domain, then the change is skin-level and is not variation of a standing-bearing constant.*

Proof. Preservation of the standing quotient, witness ledger, tensor compatibility, and continuation structure is exactly the preservation condition for admissible redescription. A difference that remains after such preservation is presentational. By Lemma 4.3, such a difference cannot be promoted into standing-bearing variation. $\square$

Lemma 5.3 (Tensor-load lemma). *If changing a purported constant changes observable compatibility, lawful continuation, standing identity, or admissible witness relations, then the change is tensor or anchor change, not free same-domain parameter variation.*

Proof. The listed changes alter the standing-bearing data of the domain. Under Definition 2.6, a same-domain variation must preserve the relevant tensor package and anchor conditions. A change in those data changes the target or violates admissibility. $\square$

Theorem 5.4 (No same-domain free constant). *No quantity can be both standing-bearing in a fixed physical domain and freely variable as a same-domain parameter of that domain.*

Proof. Assume C is standing-bearing in $(\mathcal{T}, \mathcal{A})$ and freely variable as a same-domain parameter. By Proposition 3.8, C must possess standing-preserving empirical transport in order to count as a physical constant of the domain. Since C is standing-bearing, by Lemma 5.1, alteration of C changes standing-bearing compatibility. But since the variation is asserted to be same-domain and free, it must preserve the standing quotient, witness ledger, tensor package, and admissible continuation of the domain. By Lemma 5.2, such a variation is skin and therefore not a variation of standing-bearing C . By Lemma 5.3, if it is not skin, it is tensor or anchor change and therefore not same-domain. Contradiction. Hence no such C exists. $\square$

Theorem 5.5 (Standing Fixity Theorem). *Let $(\mathcal{T}, \mathcal{A})$ be an admissibility-bearing physical domain, and let C be a physical constant used as standing-bearing load in that domain. Then C is fixed relative to the standing regime it helps constitute. Any apparent variation of C is exactly one of the following: admissible skin-level redescription, licensed continuation, boundary/envelope change, theory/domain change, or same-domain variation without preservation structure.*

Proof. By Lemma 4.2, every proposed variation falls into one of the listed role classes or lacks preservation structure. By Lemma 5.2, skin changes do not alter the standing-bearing constant. Licensed continuation does not freely vary the constant; it transports quantities according to rules already inside the domain. Boundary or envelope change alters the conditions under which the parameter is fixed. Theory/domain change changes the target. The remaining case is a same-domain value sweep over a standing-bearing quantity without preservation structure, which is excluded by Theorem 5.4. Thus the constant is fixed relative to its standing regime. $\square$

Corollary 5.6 (No inherited standing from constant-family notation). *If C is standing-bearing in a fixed domain, the notation $C \in I$, $C \mapsto C'$, or $\{C_i\}_{i \in I}$ does not by itself define admissible alternatives of the same constant. It defines either a formal family, a retyped target-domain construction if additional structure is supplied, or an unsupported inherited multiplicity.*

Proof. The notation supplies labels or coordinates, but not by itself a witness ledger, quotient identity, transport rule, boundary status, or measure. By Proposition 2.8, standing-bearing authority is not inherited by formal multiplicity alone. By Theorem 5.5, a standing-bearing constant is fixed relative to the standing regime it helps constitute. $\square$

Corollary 5.7 (Constants are not same-domain contingent explananda). *A standing-bearing constant is not a contingent explanandum within the same fixed domain in the sense required by fine-tuning parameter sweeps. It may be an empirical input, a boundary condition, or a feature to be explained by a larger theory, but not a freely variable parameter of the same standing regime.*

Proof. The only sense of contingency targeted by fine-tuning sweeps is same-domain free variation over alternative values. Theorem 5.5 excludes that sense while leaving empirical calibration, envelope inputs, and cross-theory explanation open. $\square$

6 Constants, Parameters, Units, and Running

The theorem is easily misunderstood unless several familiar parameter-types are separated.

6.1 Dimensionful constants

Dimensionful constants are not direct metaphysical explananda as bare numbers. Their numerical values depend on unit conventions. Setting $c = 1$ or $\hbar = 1$ does not change physics; it changes presentation. This does not make dimensionful constants empty. It means their standing-bearing role is relational: they fix conversion structure, symmetry structure, action/phase scales, geometric units, or other relations that remain invariant under unit redescription.

Proposition 6.1 (Unit discipline). *A dimensionful constant cannot be treated as a bare numerical target for fine-tuning until unit dependence has been quotiented away. The candidate standing-bearing content is not the raw number but the invariant relational role surviving admissible unit redescription.*

Proof. A unit change can alter the raw number assigned to a dimensionful constant while preserving all standing-bearing physical relations. By Lemma 5.2, the raw number is skin. Any load-bearing content must therefore be the invariant relation expressed by that number under a chosen unit convention. $\square$

6.2 Dimensionless constants

Dimensionless constants are more serious. A dimensionless value such as a ratio, coupling strength, angle, or mixing parameter is not removed merely by changing units. But this does not make an arbitrary alternative-values space admissible. A dimensionless quantity still needs role analysis. It may be a raw scheme coordinate, a measured boundary input, an effective parameter at a scale, a quotient-level invariant, or a standing-bearing compatibility condition.

Theorem 6.2 (Dimensionless constants still require admissible comparison). *The fact that a constant is dimensionless does not license a probability space over alternative values. A dimensionless constant may be standing-bearing, but only after the fixed-domain comparison class, witness ledger, boundary status, and measure structure have been supplied.*

Proof. Dimensionlessness removes unit dependence, not the need for standing. A value space over a dimensionless quantity still requires object identity across values, admissible transport, and a measure or ordering if probabilistic claims are made. Those structures are not supplied by the absence of units. If the quantity is standing-bearing, Theorem 5.5 excludes same-domain free variation. If it is not standing-bearing, it is not the invariant constant being explained. $\square$

6.3 Gauge couplings and RG running

Running couplings are the main physics-facing objection to a fixity claim. The answer is role discipline. A coupling value $g(\mu)$ at scale μ and in a chosen renormalization scheme is not, as a raw number, a standing-bearing invariant. It is a coordinate along a licensed continuation structure. Different schemes may use different coordinates while preserving observable content. A renormalization-group trajectory, fixed point structure, anomaly constraint, observable cross-section, or matching condition may carry invariant content; a raw coordinate value at an arbitrary scale need not.

Proposition 6.3 (RG transport proposition). *Let $g(\mu)$ be a scale-indexed coupling coordinate in a declared scheme, and suppose the fixed domain or an explicitly supplied envelope bridge licenses an RG map*

$$T_{\mu \rightarrow \nu} : g(\mu) \mapsto g(\nu)$$

through beta functions, scheme declarations, matching prescriptions, and a witness ledger. Then the indexed value together with its transport rule preserves standing under scale change. By contrast, replacing $g(\mu)$ by an arbitrary value at the same scale, without the licensed trajectory, matching data, or boundary/envelope change that would carry it, is not RG running. It is a different operation: skin reparameterization, boundary/envelope change, theory-family variation, or a same-domain sweep without preservation structure, depending on what additional structure is supplied.

Proof. An RG transformation is not a bare reassignment of a number. It is a lawful map connecting indexed quantities while preserving the observable and matching relations the effective description is licensed to carry. The transported coordinate may change, but the change is governed by $T_{\mu \rightarrow \nu}$ and by the declared scheme and witness conditions. Arbitrary reassignment lacks that transport

Quantity type	Standing role	Freely variable?	Notes
Operationally measured dimensionless ratio, e.g. mass ratios or extracted α	Witness-linked observable relation	No, once the ledger and domain are fixed	May be empirical input; alternative values require a declared comparison class.
Theoretically primitive dimensionless input, e.g. Yukawa parameter, CKM/PMNS angle, or mixing entry in a model	Boundary/envelope input or candidate tensor load	Not same-domain free	May be explained by UV structure, symmetry, or matching, but not by a sweep that lacks preservation structure.
RG-dependent effective quantity, e.g. $\alpha_s(\mu)$ or $g_i(\mu)$	Scale- and scheme-indexed continuation coordinate	Only under licensed flow	The raw number is not invariant; trajectory, fixed point, or observable matching may carry load.
Emergent low-energy fixed-point or universality datum	Tensor/anchor load within an effective domain	No within that domain	Varying it changes the universality class, cutoff envelope, or domain relation.
Cross-theory dimensionless parameter	External ensemble coordinate	Only after ensemble structure	Requires object-class, measure, and witness bridge before probability claims are physical.

Table 2: Dimensionless constants require role typing before variation claims.

relation. Therefore RG evolution is a standing-preserving continuation internal to, or explicitly bridged between, admissible envelopes; arbitrary value replacement is not. $\square$

Corollary 6.4 (Running does not refute fixity). *Renormalization-group running does not show that standing-bearing constants are freely variable. It shows that some printed parameters are continuation coordinates whose invariant content, if any, lies in the licensed trajectory, matching data, fixed point, observable relation, or boundary/envelope input.*

6.4 Effective-field-theory parameters

Effective parameters are envelope-bounded. They are meaningful inside a domain with cutoffs, symmetries, matching conditions, and experimental reach. Treating them as if they were unrestricted fundamental constants is a scope error. Conversely, varying them inside an EFT may define a family of models; but the family becomes a physical explanatory space only after its admissibility envelope, matching conditions, measure, and witness ledger are specified. This is fully compatible with ordinary EFT methodology: the theorem constrains only the interpretation of parameter variation as an explanation of standing-bearing constants, not the use of EFT coefficients inside their declared range of validity.

Proposition 6.5 (Envelope boundedness). *An effective parameter is fixed only relative to the envelope in which it is used. Moving it beyond that envelope without supplying new matching and*

witness structure does not by itself explain the original standing-bearing constant; it changes or extends the transport problem.

Proof. The standing of an effective parameter depends on the domain where the effective description is valid. Changing the cutoff, degrees of freedom, matching condition, or regime changes the comparison class. If those changes are licensed, the parameter has been transported by new structure. If not, the parameter has been moved outside its admissibility envelope without the bridge needed for same-object comparison. $\square$

7 Parameter-Illusion Classification

The following classification is the practical center of the analysis. It determines what an apparent constant-variation claim is actually doing.

Table 3: Classification of apparent constant variation.

Route	Description	Status under fixity
Unit normalization	Raw number changes because units change; invariant relations remain.	Skin; not variation of standing-bearing content.
Chart/scheme reparameterization	A coordinate transformation changes the printed parameter while preserving observables or compatibility.	Skin if invertible and role-preserving.
Gauge or representation display	Same physical content is expressed in different gauge or representation.	Skin if quotient and tensor content are preserved.
Boundary/envelope input	A measured value, cutoff, matching condition, initial condition, or calibration fixes the parameter in a domain.	Legitimate input; not a same-domain free variable.
Licensed continuation	Scale/time/RG evolution changes indexed values according to domain laws.	Continuation, not free variation.
Effective-model family	A family of models varies coefficients inside an explicitly declared envelope.	Legitimate model family if witness and measure are supplied; not automatic explanation of constants.
Theory/domain change	Altering the value changes symmetry, field content, consistency, observables, or identity conditions.	New domain or theory; not same-domain variation.
Cross-domain ensemble	A landscape/multiverse supplies multiple domains with comparison rules.	Possible only with added ensemble, measure, and witness ledger.
Same-domain value sweep	Values are swept while the original standing regime is assumed to remain comparable and unchanged.	Requires preservation structure; otherwise not admissible for standing-bearing constants.

Theorem 7.1 (Parameter-illusion exhaustion). *Every apparent variation of a physical constant falls into one of the routes in Table 3. Only the same-domain value sweep is directly excluded; the others are reclassified according to role.*

Proof. Let a proposed variation of C be given. If all invariant content is preserved, the variation is unit/chart/gauge/scheme skin. If the value changes under a licensed flow, it is continuation. If calibration, cutoff, matching, or boundary data are changed, it is boundary/envelope variation. If the physical object-class, consistency conditions, observables, field content, or standing quotient change, it is theory/domain change. If a plurality of domains is asserted, a cross-domain ensemble has been introduced. The only remaining case is a value sweep that preserves none of these licenses while asserting same-domain comparability. That is the unsupported case excluded by Theorem 5.4. $\square$

Corollary 7.2 (Correct target of the paper). *The paper does not reject model-building with parameters. It rejects only the promotion of a numerical sweep whose standing structure has not been supplied into a physically admissible explanation of standing-bearing constants.*

7.1 Constant families and explicitly governed scans

The classification in Table 3 separates three cases that are often conflated. A *formal* value-family is notation only. An *inherited* value-family treats that notation as if standing, comparability, and measure authority automatically transfer from the original domain. An *explicitly gated* value-family supplies a new target domain with declared transport, witness, envelope, and comparison structure. The first is harmless bookkeeping, the second is the target of the Fixity theorem, and the third remains admissible as a model-family or ensemble only under its own gate.

This distinction protects ordinary physics practice. Controlled parameter scans, EFT coefficient families, matching studies, and UV-completion searches may be legitimate within explicitly governed target domains. They become problematic only when the family is treated as an inherited same-domain explanation of a standing-bearing constant without supplying the target-domain structure that makes comparison meaningful.

8 Fine-Tuning and the No-Measure Result

Fine-tuning arguments typically require at least four objects:

- (F1) a parameter space $\mathcal{P}$ of alternative constant assignments;
- (F2) a subset $L \subseteq \mathcal{P}$ counted as life-permitting, structure-permitting, observer-permitting, or otherwise target-permitting;
- (F3) a measure μ or probability distribution over $\mathcal{P}$;
- (F4) an identity-transport rule saying that the points of $\mathcal{P}$ are comparable alternatives of the same explanatory object.

The problem is not merely that μ is hard to choose. The stronger problem is that $\mathcal{P}$ itself is not a physical object until its standing conditions are fixed. The target is not every useful model-dependent measure; it is the claim that a canonically privileged standing-preserving measure is already available over a pre-admissible value space without further structure.

Definition 8.1 (Admissible parameter space). *A parameter space $\mathcal{P}_C$ for a constant C is admissible only if it supplies:*

- (P1) *a fixed object-class for what varies;*
- (P2) *a standing quotient identifying when two presentations are the same physical content;*

- (P3) a witness ledger and transport rule comparing points of the space;
- (P4) boundary/envelope rules governing which points remain in the same domain;
- (P5) a measure, metric, ranking, or typicality structure if probabilistic claims are made.

Theorem 8.2 (No canonical pre-admissible measure). *Before a constant-variation space is made admissible in the sense of Definition 8.1, there is no canonically privileged standing-preserving metric, measure, probability distribution, typicality ordering, or representative over its values.*

Proof. A metric or measure over values requires determinate objects to be compared and a rule assigning distances, volumes, weights, or representatives. Prior to admissible fixation, the alleged values have no fixed object-class, no standing quotient, no witness ledger, and no transport rule. Any measure imposed at that stage is therefore not extracted from the physical domain as a canonically privileged standing-preserving measure. It may be a useful modeling convention, an explicitly added prior, a coordinate artifact, or an external selector. Such a device may be studied once declared as part of a new model, but then the model has added structure; the measure was not supplied by the original pre-admissible value space. $\square$

Corollary 8.3 (Fine-tuning is conditional, not primitive). *A fine-tuning probability has canonical explanatory force only conditional on an already admissible parameter space and a declared standing-preserving measure. It does not by itself establish the admissibility of the parameter space and measure on which it depends.*

Proof. The probability calculation presupposes $\mathcal{P}$, μ , and cross-point comparison. By Theorem 8.2, these are not canonically supplied before admissible fixation. Hence a fine-tuning probability may be a conditional result inside a specified ensemble, but not a primitive explanation of why the constants have standing. $\square$

8.1 What remains of fine-tuning

The result does not discard all fine-tuning language. Fine-tuning can remain a heuristic marker that a theory is sensitive to boundary inputs, that a model lacks a known dynamical account, or that an ensemble hypothesis would need a sharply constrained measure. But the strong explanatory claim becomes conditional: one should not assign canonical explanatory authority to claims of surprise until the prior and the space on which it lives have been admitted as standing-preserving structure.

A coordinate interval such as $\alpha \in (0, 1)$ is not yet a physical sample space. A uniform prior over α is not invariant under $\alpha \mapsto \log \alpha$, $\alpha \mapsto \alpha^2$, or other reparameterizations unless extra structure is supplied. The point is not that such priors are unusable as modeling conventions; it is that their canonical explanatory authority is underdetermined until the admissible measure and comparison structure are declared.

For example, a scan over the fine-structure constant, the Higgs vacuum expectation value, the QCD scale, or CKM angles is not automatically one type of operation. It may be a unit-independent ratio scan, a boundary-input scan, an EFT matching scan, a UV-completion hypothesis, or a same-domain sweep without preservation structure. Fine-tuning arguments become physically sharp only after that distinction is made. Appendix A illustrates the classification in the special case of the fine-structure constant.

9 Anthropic Reasoning under Standing Constraints

Anthropic reasoning is often invoked as a response to fine-tuning. The standing-fixity result does not rule out anthropic reasoning. It constrains its legitimate order of operation.

Definition 9.1 (Admissible anthropic conditional). *An anthropic conditional is admissible only when it operates over a standing-preserving ensemble whose members, observer conditions, witness roles, and measure structure are already fixed independently of the anthropic conditioning step.*

Theorem 9.2 (Anthropic selection is conditional on a fixed constant-space). *Anthropic reasoning does not by itself explain standing-bearing constants when it presupposes the cross-domain parameter space, measure, and identity transport that constant-variation fails to preserve or leaves unfixed.*

Proof. Anthropic conditioning selects among already defined alternatives. To condition on the existence of observers across varying constants, the alternatives must already be individuated, comparable, and measurable. But if the varied constants are standing-bearing, Theorem 5.5 says that same-domain free variation lacks preservation structure; if the alternatives are cross-domain, Definition 8.1 requires additional ensemble and transport structure. The anthropic step does not supply this structure without circularity, because it uses the structure to state the condition. $\square$

Corollary 9.3 (Anthropic reasoning as posterior filter). *Anthropic reasoning may function as a posterior filter inside a licensed ensemble. It does not function as the source of admissibility for the ensemble, measure, or constant-comparison class.*

The distinction between inherited and explicitly gated constant-families is decisive here. A life-permitting region inside a declared target-domain model may be studied conditionally. What does not follow is that observer compatibility in one domain automatically transports standing across a family of alternative constant assignments. Observer predicates, measures, and cross-domain comparison inherit no authority merely from the written multiplicity of constants.

For the purposes of the present paper, anthropic reasoning is evaluated only as a proposed explanation of the fixed-domain standing of a constant. It is not treated as an independent uniqueness thesis; its relevance is whether the constant-continuation locus, witness ledger, and ensemble bridge required by the fixity target have been supplied before selection is invoked.

9.1 The life-permitting subset problem

Even after an ensemble is specified, the subset called “life-permitting” or “observer-permitting” requires its own role discipline. A change in constants may alter chemistry, stellar structure, cosmological expansion, biological pathways, or the meaning of observation itself. The anthropic condition therefore cannot be treated as a simple subset of a pre-given Euclidean parameter space unless the relevant identity and witness conditions are held fixed. Otherwise the argument compares different object-classes while pretending to sample one object.

10 Multiverse, Landscape, and Everettian Non-Branching

Multiverse and landscape explanations are not identical to fine-tuning arguments. They may attempt to supply a real ensemble rather than merely imagine a parameter interval. The standing-fixity constraint is correspondingly conditional: if the ensemble supplies admissible structure, it can be studied; if it does not, its explanatory force remains underdetermined.

The fixity question remains prior in either case: the model must specify which constant-continuation loci are preserved, which are retyped as target-domain parameters, and which are replaced by a different standing regime. Without that role assignment, the ensemble is not yet explaining the same standing-bearing constant.

Definition 10.1 (Constant-varying ensemble). *A constant-varying ensemble is admissible only if it supplies:*

- (E1) *an object-class for ensemble members;*
- (E2) *a rule connecting member-level constants to common explanatory terms;*
- (E3) *a measure or weighting rule, if statistical explanation is claimed;*
- (E4) *a witness ledger or observational bridge connecting ensemble claims to empirical use;*
- (E5) *a boundary rule distinguishing domain variation from mere redescription.*

Theorem 10.2 (Landscape non-automaticity). *A landscape or multiverse proposal does not automatically explain constants merely by containing multiple values. It explains them only to the extent that it supplies admissible ensemble structure, measure, and witness transport rather than presupposing them.*

Proof. Multiple values without object-class, measure, and witness transport are merely a collection of descriptions. A statistical or anthropic explanation requires comparison across the collection. By Definition 10.1, that comparison needs additional structure. If the structure is supplied, the explanation is conditional on it. If not, the proposal repeats the same pre-admissible parameter-space error identified in Theorem 8.2. $\square$

Definition 10.3 (Same-standing and cross-standing ensembles). *A same-standing ensemble is an ensemble whose members preserve the object-class, admissibility envelope, witness ledger, transport rules, and tensor/anchor role assignments of the target domain. A cross-standing ensemble is an ensemble whose members differ in at least one of these standing conditions.*

Proposition 10.4 (Ensemble standing dichotomy). *A same-standing ensemble does not support free scanning of standing-bearing constants, while a cross-standing ensemble does not inherit same-domain probability, typicality, or identity transport without additional bridging structure.*

Proof. In a same-standing ensemble, standing-bearing constants remain part of the preserved tensor/anchor package. Theorem 5.4 therefore excludes free same-domain variation of those constants. In a cross-standing ensemble, the variation occurs across domains, theory-families, boundary envelopes, or witness ledgers. Such variation may be legitimate, but it no longer inherits same-domain identity or measure. It needs an explicit bridge specifying what is being compared, how ensemble members are weighted, and how claims about them return to empirical use. $\square$

This distinction prevents two opposite errors. The paper does not dismiss all multiverse or landscape models merely because they involve different constants. But it also does not grant that a family of different EFT vacua, boundary conditions, compactifications, or low-energy parameter sets automatically possesses a physically canonical measure over “the same” constants. Same-standing variation is too restrictive to allow free standing-bearing scans; cross-standing variation is permissive only after the ensemble bridge has been supplied.

When an ensemble theory succeeds in supplying membership, transport, witness, and measure structure, the result is best treated as a newly typed target domain. The constants of the original domain are then not freely varied inside that original domain; they are compared through a further construction whose own admissibility must be evaluated. This retyping preserves legitimate ensemble modeling while blocking inherited standing-bearing multiplicity from the original constant notation alone.

10.1 Everettian branching

Everettian branching is especially clear. In the orthodox Everettian picture, branches arise from unitary evolution of a fixed quantum state under a fixed Hamiltonian or law-structure, with branch-relative records inheriting that shared dynamical package. Branches differ by outcome and record structure under the fixed dynamics; they do not, by ordinary branching alone, differ in the constants defining that dynamics.

Theorem 10.5 (Everettian non-branching of physical constants). *In an Everettian framework whose branches arise by unitary evolution under a fixed physical law, physical constants do not vary across branches as ordinary branch variation. Constant variation would constitute law, Hamiltonian, boundary, or theory-family variation rather than outcome branching.*

Proof. Everettian branches are branches of a fixed dynamical package: the Hamiltonian, Hilbert-space structure, state evolution rule, and semantic interpretation that make branch records identifiable. The constants appearing in that package are part of the law and standing structure under which branch decomposition is defined. If a branch is assigned different constants, the law generating the branch structure has changed or a meta-ensemble of laws has been added. That is not ordinary branching within the original Everettian domain. It is theory/domain variation requiring additional ensemble structure. $\square$

Corollary 10.6 (No anthropic branching shortcut). *Everettian branching does not by itself license anthropic scanning over physical constants. Any such scanning requires a further law-varying ensemble or meta-theory.*

11 Null-Regime and No-Default-Selector Closure

A remaining escape route is to imagine that constants are chosen from a lower, prior, neutral, or null regime. The null-regime and constraint-bundle material rules out this move in the limited form needed here.

Definition 11.1 (Null selector). *A null selector is a purported rule that selects, ranks, measures, or validates constant-values while claiming not to add admissible structure, witness authority, metric structure, boundary conditions, or standing-bearing content.*

Lemma 11.2 (Null inertness). *A purportedly null selector has no selective authority. If it distinguishes one constant-value, measure, or representative from another, it is not null; it has added structure. If it adds no structure, it supplies no distinction.*

Proof. Selection requires a distinction that is retained as relevant. Relevance requires role-bearing structure: a metric, ordering, witness, boundary, or rule. A genuinely null regime supplies none of these. Therefore it has no selective role. If a distinction is made, the selector is no longer null and must be admitted as added structure. $\square$

Theorem 11.3 (No default parameter space). *There is no default admissible parameter space of physical constants beneath or prior to the standing regime of a physical domain. Any such space is either AMetric and inert, or it imports metric, measure, selector, boundary, or witness structure.*

Proof. A default parameter space capable of explanatory work must support at least comparison and, for fine-tuning, measure. If it supplies no structure, Lemma 11.2 makes it inert. If it supplies structure, the structure must be identified and licensed; it is no longer default or null. Thus no lower parameter space can govern constants without becoming an explicit added domain. $\square$

11.1 Positive closure: non-vacuous governance

The positive constraint-bundle point is equally important. Once non-degenerate physical reference is in force, there is already non-vacuous invariant constraint architecture: identity fixation, witness role, boundary typing, transport, quotient stability, and fail-closed treatment of standing failure. Erasing this architecture does not explain constants, because it also erases the conditions under which constants can be physically referred to. An arbitrary selector outside the architecture likewise functions as an added primitive rather than as an explanation internal to the fixed domain.

The conclusion is not an ontological thesis about necessary realization. It is narrower: once non-degenerate standing-preserving physical reference is in force, constants are not governed, within that reference structure, by total vacuity, arbitrary selection, or a pre-physical default measure. Their explanatory treatment must respect the standing architecture already required for physical reference.

12 Objection Matrix

Table 4: Objections and replies.

Objection	Reply
You did not derive the constants.	Correct. The theorem is not a numerical derivation. It is a role theorem: same-domain free variation of standing-bearing constants lacks preservation structure. Numerical derivation would require additional dynamics, boundary conditions, or a larger theory.
This is just conventionalism.	No. The paper separates convention from invariant load. Unit choices, charts, and schemes are conventional skin; observable compatibility, witness transport, and standing identity are not. Conventionalism is rejected exactly where skin is promoted into tensor authority.
Constants can vary in other theories.	Yes. Other theories may have different constants. That is theory/domain variation, not free variation of the same standing-bearing constant inside a fixed domain. The paper permits this and requires it to be typed honestly.
Couplings run.	Running couplings are licensed continuation coordinates or effective parameters. Their running shows that raw numeric values are often not standing-bearing constants. It does not show that standing-bearing invariant load is freely variable same-domain.

Objection	Reply
Anthropic reasoning is still possible.	It is possible as a posterior filter over a licensed ensemble. It does not function as the source of the ensemble’s standing, measure, or cross-domain identity transport.
Multiverse models can define ensembles.	They can. The paper’s claim is conditional: if the ensemble, measure, and witness bridge are supplied, the model can be studied. If they are merely assumed by drawing a parameter space, the explanation is unsupported.
The theorem is metaphysical necessity in disguise.	No. The theorem is relative to a fixed admissibility-bearing domain. It does not rank all possible worlds or deny that different domains may instantiate different laws.
This restricts physics.	It constrains only explanatory shortcuts whose comparison structure is not supplied. It leaves empirical measurement, EFT modeling, RG analysis, symmetry constraints, BSM theory-building, and ensemble construction available when their domains and witnesses are declared.
Fine-tuning is empirically useful.	It may be heuristically useful. The theorem targets the stronger claim that a fine-tuning probability over constants has canonical explanatory authority before a declared measure, admissible comparison class, and standing-preserving transport are supplied.
The life-permitting range is obvious enough.	Some dependencies may be robust, but a life-permitting subset is still a subset of a space. The space, transport, and measure must be fixed before probability or typicality claims follow.
Standing is just stipulated.	No. Proposition 3.8 identifies standing as the empirical transport condition required for a parameter to be used across observers, calibrations, experiments, and lawful redescriptions. Without that transport, the alleged constant is not a stable physical object.
What about different EFT vacua or compactifications?	They are cross-standing ensemble candidates. They may be studied if a bridge, measure, and witness relation are supplied, but they do not inherit same-domain constant identity automatically.

13 Consequences for Theory Construction

The Fixity theorem changes how constants should be handled in foundational work.

13.1 A classification program

Rather than asking immediately why a printed constant has a value, one should first classify the role of the printed quantity:

(K1) Is it dimensionful skin or invariant relational load?

(K2) Is it a chart/scheme coordinate or redescription-invariant content?

(K3) Is it a boundary/envelope input fixed by measurement or matching?

- (K4) Is it an effective parameter whose standing is bounded by a cutoff or regime?
- (K5) Is it a licensed continuation value indexed by scale or time?
- (K6) Is it a tensor/anchor condition whose alteration changes the domain?
- (K7) Is a proposed alternative-values space equipped with an admissible measure and witness ledger?
- (K8) Is the proposed family merely formal, inherited without standing structure, or explicitly gated as a new target domain?

Only after this classification is complete can an explanation of a constant be typed correctly.

13.2 What can be targeted next

The natural next technical program is not to claim that all constants are numerically necessary. It is to classify constants by role:

- (M1) **Standing-fixed invariants:** quantities whose alteration changes the standing regime.
- (M2) **Envelope-fixed inputs:** quantities fixed by boundary, calibration, matching, or empirical ledger.
- (M3) **Trajectory-fixed quantities:** quantities fixed by licensed continuation or RG flow.
- (M4) **Skin-level labels:** quantities whose raw numerical form changes under admissible redescription.
- (M5) **Cross-domain variables:** quantities meaningful only inside an explicitly supplied ensemble or theory-family.
- (M6) **Constant-continuation loci:** preserved standing roles that identify what remains the same through redescription, calibration, and transport.

This classification is the physics-facing extension of standing fixity.

14 Conclusion

The apparent mystery of physical constants is often framed as a problem of improbable values. That framing assumes a parameter space of alternatives with a measure, a typicality standard, and cross-realization identity. The present paper argues that these assumptions are not free.

Within a fixed admissibility-bearing physical domain, a standing-bearing physical constant does not function as a freely variable same-object parameter. If an apparent variation preserves all invariant content, it is skin. If it changes invariant content, it changes tensor, anchor, envelope, or domain. If it invokes a prior probability over values, it must first provide an admissible measure. If it invokes anthropic selection, it must first provide a standing-preserving ensemble. If it invokes Everettian branching, it must not confuse outcome branching under fixed law with law variation. In each case, the issue remains the same: whether the proposed explanation preserves the standing-bearing identity of the constant being explained.

The result does not answer the numerical question “why this value?” by deriving the value. It clarifies the prior question: what would count as an admissible explanation of a constant at all?

The answer is fixed-domain role discipline. Physical constants therefore do not function as freely variable same-domain explananda within standing-preserving physical description. Depending on role, they are standing-fixed invariants, envelope inputs, continuation coordinates, or skin-level labels. Treating them as freely sampled alternatives before that role is fixed does not yet constitute a physical explanation; it is a value space awaiting admissible comparison structure.

The positive lesson is that constants need not be treated as arbitrary numerical accidents awaiting external selection. They can be analyzed as invariant structural components participating in the conditions for stable physical reference, empirical transport, and domain-bounded theory application. Formal value-families, effective parameter scans, and ensemble constructions remain available when they are explicitly gated; what is rejected is inherited standing-bearing authority from multiplicity alone. This reframing does not end the physics of constants. It clarifies which explanatory moves preserve the standing of the object being explained while leaving ordinary empirical calibration, EFT practice, RG transport, controlled parameter-family modeling, and explicitly declared ensemble construction intact.

A Illustrative Standing Classification for the Fine-Structure Constant

This appendix gives a compact worked illustration of the role discipline used in the paper. It is not a numerical derivation of the fine-structure constant. It shows how the same printed symbol can participate in several distinct operations, only some of which count as physically contentful variation of a standing-bearing constant.

In SI notation one often writes

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}.$$

In natural or rationalized unit conventions the displayed formula changes. The unit-sensitive pieces e , ϵ_0 , $\hbar$, and c may be redistributed, normalized, or absorbed into conventions. This is not a change of the electromagnetic standing relation measured by α ; it is a change of skin. The dimensionless relation survives the unit redescription.

A second operation is scheme or scale presentation. In quantum field theory, the electromagnetic coupling is represented by a scale-dependent effective quantity once renormalization and matching conventions are specified. Such running is not a free same-domain variation. By Proposition 6.3, it is licensed continuation: a value at one scale is connected to a value at another scale by the theory’s renormalization prescription, scheme declaration, and empirical matching rules. The standing-bearing object is not the isolated printed number alone, but the transported relation among scale, scheme, witness ledger, and observable predictions.

A third operation is empirical calibration. A measured low-energy value of α can be treated as an envelope or boundary input for a model. The model may not derive that value internally, but once the value is admitted as calibration data it has standing inside the domain. Treating that calibrated input as unexplained is different from treating it as freely variable while preserving the same domain.

A fourth operation is a counterfactual sweep without preservation structure, for example writing $\alpha \in (0, 1)$ and assigning a uniform prior to the interval. That operation is not yet a physical fine-tuning analysis. The interval does not by itself specify the object-class being varied, the theory-family in which the variation occurs, the witness ledger comparing worlds or models, the boundary rules preserving domain identity, or a coordinate-independent measure. A prior over

the printed coordinate may be an explicitly declared modeling convention; however, it is not a canonically privileged standing-preserving measure supplied by the bare notation $\alpha \in (0, 1)$.

A fifth operation is an explicitly governed target-domain study, such as a specified EFT family or UV-completion class with declared matching rules and observables. That operation is not rejected by standing fixity. It is simply no longer an inherited same-domain variation of the original constant; it is a controlled comparison inside a further admissibility-bearing construction.

Operation involving α	Role classification	Standing-fixity reading
Change units or rationalization convention	Skin redescription	The printed formula changes while the dimensionless electromagnetic relation is preserved.
Change renormalization scale or scheme under a declared prescription	Licensed continuation plus scheme skin	The isolated number changes, but the change is transported by RG and matching structure.
Use a measured value as an input to a bounded model	Boundary/envelope input	The value is fixed for that model once calibrated, though not necessarily derived by the model.
Compare different UV completions or EFT vacua with different low-energy α	Cross-standing or theory-family ensemble	Legitimate only with explicit bridge, witness ledger, measure, and target-domain gate.
Sweep α freely over a coordinate interval inside the same standing domain	Same-domain sweep without preservation structure	Requires object identity, transport, and measure.

Table 5: Illustrative role classification for the fine-structure constant.

This example shows why the Fixity theorem does not deny unit changes, RG running, empirical calibration, or theory-family comparison. It rejects only the inference from a writable coordinate interval to a standing-preserving sample space over freely variable same-domain constants.

B Dependency Map to the Companion Corpus

The proof structure above imports, in compressed form, several companion results from the AASC and constraint-bundle/null-regime corpus. The imports are role-limited:

Companion result	Use in this manuscript
The Structure of Admissibility	Provides fixed-domain standing conservation, bivalence/fail-closed discipline, AMetric boundary posture, and no-repair treatment of standing failure.
Standing-Admissibility Identity Closure	Supports Lemma 3.10: standing and admissibility cannot be separated by a second same-domain classifier.
Anchor, Tensor, and Skin	Supplies the role decomposition used in Section 4; in particular, skin cannot carry gate, selector, or tensor authority.
Gauge-Coupling Parameters and Forced Uniqueness	Supplies the gauge/RG role discipline used in Proposition 6.3: raw chart-numeric couplings are not standing-bearing as raw numbers, while invariant trajectory or overlap content may be tensor-eligible.

Companion result	Use in this manuscript
A Classification Theorem for Admissible Standard-Model Structures	Supplies the caution that envelope inputs and admissible bookkeeping occupants are not automatically numerical derivations of all parameters.
The AMetric Boundary	Supplies Theorem 8.2: before admissibility is fixed, there is no canonically privileged standing-preserving metric, measure, probability, typicality order, selector, or representative.
Admissible Record Construction in Physically Non-Selective Quantum Regimes	Supplies the Everettian role distinction: branch-relative structure under fixed law does not by itself create cross-law constant variation.
The Structural Inadmissibility of the Null Regime	Supplies null inertness: total vacuity does not function as selector, endpoint, lower generator, or same-type counterexample inside admissibility-bearing evaluability.
Constraint-Bundle Forcing under Fixed-Domain Evaluability	Supplies the positive closure that non-degenerate evaluability forces non-vacuous invariant role-constraint architecture.
Concrete-Domain Closure under AASC	Supplies the non-dischargeability posture: realized persistence and bounded standing cannot be replaced by abstract-only, skin-only, or AMetric closures.
Continuation-Locus Induction	Supplies the persistence discipline behind constant-continuation loci: non-trace continuation requires a preserved admissibility-bearing locus up to lawful redescription.
Transport-Closure Locus for Continuous Parameters	Supplies the constraint that continuous parameter families must close under admissible transport or quotient into invariant classes rather than inherit standing automatically.
Generation Multiplicity Minimality and Non-Degeneracy	Supplies redundancy and unbounded-replication discipline for standing-indistinguishable replicated sectors.
Ensemble Embedding Collapse under Standing-Preserving Admissibility	Supplies the retyping rule distinguishing formal multiplicity, inherited standing-bearing multiplicity, and explicitly gated target-domain constructions.

C Compact Formal Summary

Let $D = (\mathcal{T}, \mathcal{A})$ be a fixed physical domain. Let $\sim_{\mathcal{A}}$ be admissible-redescription equivalence on presentations, and let $[x]_{\mathcal{A}}$ denote a standing identity class. Let C be a candidate constant-role.

Definition C.1 (Role map). *A role map is a partial assignment*

$$\rho_D : \{\text{descriptive elements of } D\} \rightarrow \{\text{Anchor, Tensor, Skin, Envelope, Continuation}\}$$

licensed only when the assigned role preserves the standing quotient and witness ledger of D .

Definition C.2 (Formal target-domain retyping). *A family $\{D_i\}_{i \in I}$ is an explicitly gated target-domain construction only when it includes its own admissibility predicate, witness ledger, standing quotient, transport relation, envelope data, and any measure or selector it uses. Otherwise it is a formal or inherited family, not an admissible same-domain comparison space.*

Definition C.3 (Free same-domain variation). *A free same-domain variation of C is a family $\{D_c\}_{c \in I}$ such that:*

- (V1) D_c is asserted to remain the same standing domain as D for all $c \in I$;
- (V2) C takes distinct values across I ;
- (V3) no new boundary, witness, transport, measure, or theory-family structure is added;
- (V4) the variation is claimed to be physically contentful rather than skin.

Proposition C.4 (Formal no-free-constant statement). *If C is standing-bearing in D , then no free same-domain variation of C exists.*

Proof. If the variation preserves $[x]_{\mathcal{A}}$, witness compatibility, and tensor compatibility for all relevant presentations, then it is skin, contradicting (V4). If it changes any of those structures, then D_c is not the same standing domain as D unless additional transport or boundary structure is supplied, contradicting (V1) or (V3). Therefore no family satisfies all four clauses. $\square$

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